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Creating Custom Security Roles in D365

Document Version Control

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V.0.1	05-06-2025	Shamsher Singh	Initial Draft

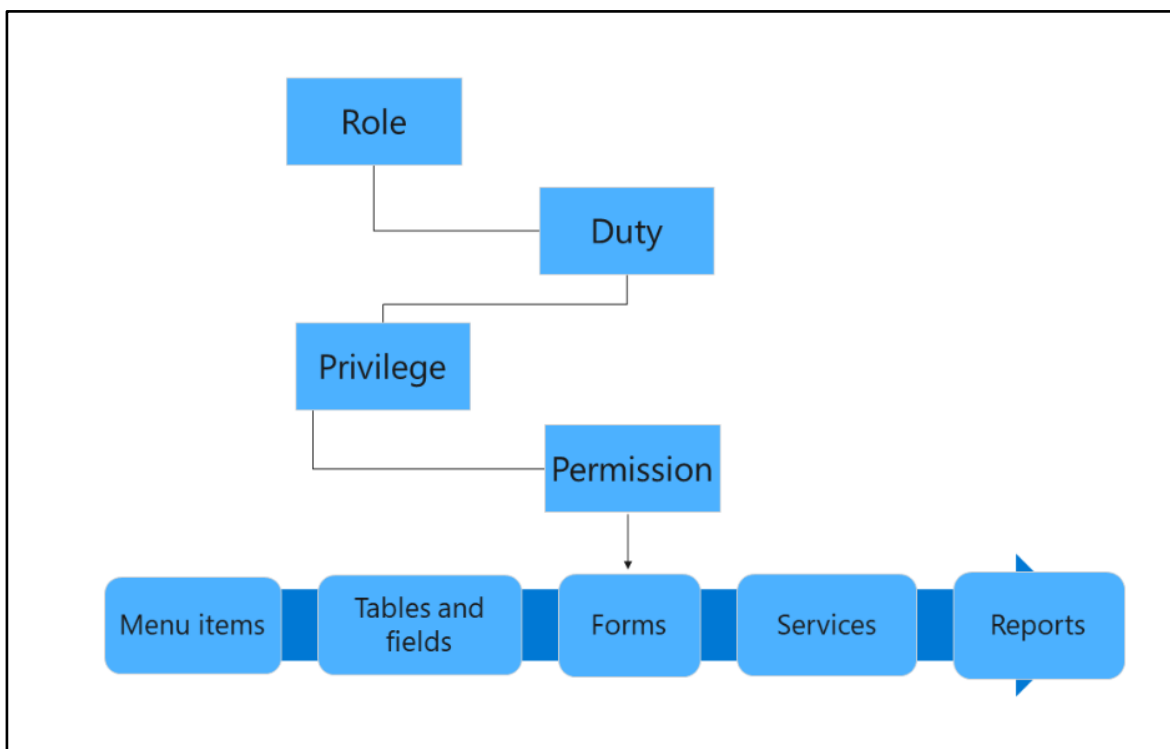
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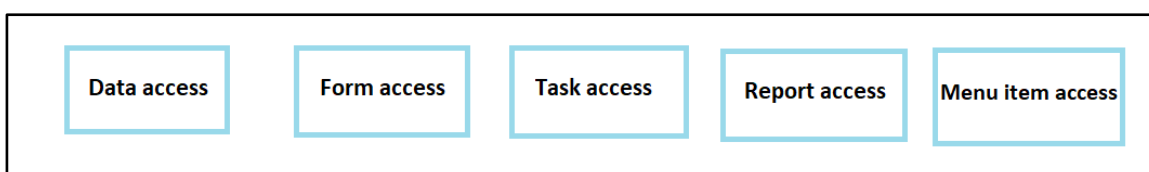
1 Understanding D365 Security Architecture

In this document we will walk through the process of creating a Custom/view-only security role in Microsoft Dynamics 365 Finance & Operations (D365F&O), using role-based security configuration. Additionally, we'll demonstrate how to assign this role on a legal entity basis. While it's possible to customize existing security roles, for this guide, we'll focus on creating a new role from scratch and assigning it view-only permissions. The same approach can be applied to any form within the system, allowing you to either create new roles or assign multiple permissions to existing duties.

Before diving into role creation, it's important to understand the security architecture of D365F&O. This foundational knowledge makes it significantly easier to tailor security configurations to meet specific business needs. Below is a high-level overview of the security model, which includes roles, duties, and privileges and how they relate to each other:



With role-based security, administrators can assign specific roles to users and then assign permissions to those roles. The permissions that can be assigned include:



1.1 Key Concepts in D365 Security

Microsoft Dynamics 365 leverages a comprehensive, role-based security model to manage user access and protect sensitive data. Hosted within Microsoft Azure's secure and scalable infrastructure, it ensures high performance, reliability, and enterprise-grade security.

This security framework is built on the principle of least privilege—granting users only the access necessary to perform their assigned duties. Through clearly defined roles, privileges, and hierarchical structures, Dynamics 365 enforces data integrity and minimizes security risks.

Designed for modern organizations, the model supports compliance, enhances operational control, and ensures users interact with the system in a secure, controlled manner.

1.2 Security Roles

A security role in Microsoft Dynamics 365 is a structured set of privileges that governs what actions a user can perform and which data they can access within the system. These roles are assigned to individual users or teams and are essential in enforcing access control and operational boundaries.

For example:

- Sales Manager: Has the ability to view and edit all sales-related records across the organization.
- Customer Service Representative: Restricted to viewing and editing only the cases assigned specifically to them.

This role-based approach ensures that users have the appropriate level of access based on their responsibilities, supporting both security and efficiency.

1.3 Duties

A duty is a logical grouping of privileges that represents a specific business task or responsibility within Microsoft Dynamics 365. Duties are modular and reusable, allowing them to be associated with multiple security roles to streamline role configuration and management.

For example:

- Maintain Customer Records: Grants privileges to create, read, update, and delete customer data.
- Generate Invoices: Grants privileges to create and post invoices.

By organizing privileges into duties, organizations can ensure consistency, simplify role maintenance, and enhance the security model's scalability.

1.4 Privileges

A privilege represents the most granular level of access control in Microsoft Dynamics 365. It defines a user's ability to perform a specific action—such as creating, reading, updating, or deleting—on a particular type of data record.

Privileges are the foundational components of the security model. They are grouped into duties, which are then assigned to roles, forming a structured hierarchy of access.

Examples of privileges include:

- **Read:** Allows viewing of a record.
- **Write:** Allows editing of a record.
- **Delete:** Allows removal of a record.

This layered structure ensures precise access control while maintaining flexibility and consistency across roles.

1.5 Uses of Roles, Duties and Privileges works in D365

The Microsoft Dynamics 365 security model is built on a **hierarchical structure** that promotes flexibility, consistency, and reusability across the organization:

- **Privileges** define specific actions on data (e.g., Read, Write, Delete).
- **Duties** group related privileges to represent a task or responsibility.
- **Roles** consist of one or more duties and define access at a functional level.
- **Users or Teams** are assigned roles to control what they can access and do in the system.

For example, the **Maintain Customer Records** duty—comprising privileges to create, read, update, and delete customer records—can be included in both the **Sales Manager** and **Customer Service Representative** roles. This ensures consistent access rules while allowing roles to be tailored to different business functions.

1.6 Designing Security roles in d365

As a consultant, building a secure and scalable security model involves a structured approach. Follow these key steps to design effective security roles that align with organizational needs while maintaining data protection:

1. **Analyze Business Requirements**
 - Identify all user groups (e.g., Sales, Marketing, Finance).
 - Understand the tasks each group performs and the data they require access to.
2. **Define Privileges**
 - Determine specific actions each group must perform (e.g., Read, Write, Delete).
 - Ensure alignment with the **principle of least privilege**, granting only necessary access.

3. Group Privileges into Duties

- Create duties that represent business tasks or responsibilities.
- *Example:* A **Generate Reports** duty might include privileges to read data and run reports.

4. Create Security Roles

- Combine duties into roles based on job functions.
- *Example:* A **Sales Representative** role might include **Maintain Customer Records** and **Generate Invoices** duties.

5. Assign Roles to Users or Teams

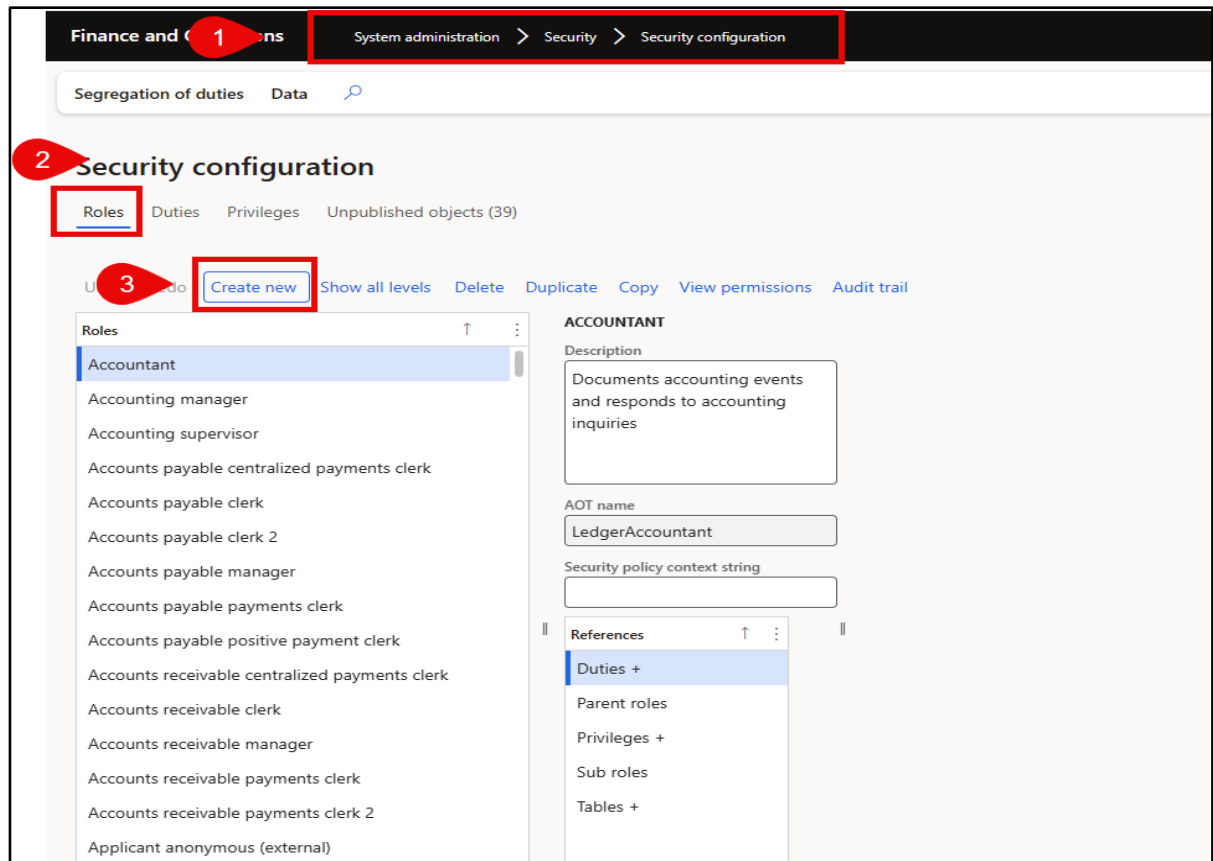
- Assign roles based on the user's function in the organization.
- Use **teams** to manage access consistently across groups, simplifying role administration.

This systematic approach ensures secure, role-based access control while supporting operational efficiency and compliance.

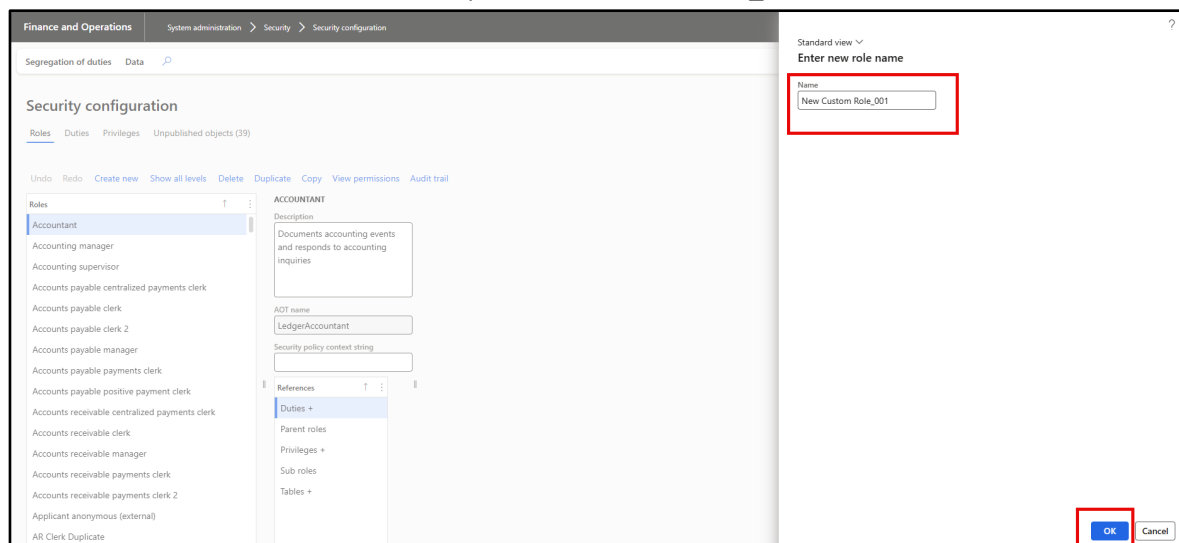
2 Create New Custom Security Role

Scenario: We will create a new 'view only' security role for the Customer master data. We can apply the same process on any D365 form.

1. Go to System Administrator > Security > Security Configuration > Role, then click Create New to define and save a new security role.



2. Define the name for the role, for example: "New Custom Role_001"



3. Create a new duty by navigating to **Duties > Create New**, enter the name "New Custom Duty_001", then click **OK**.

The screenshot shows the 'Security configuration' window with the 'Duties' tab selected. A red circle 1 points to the 'System administration > Security > Security configuration' breadcrumb. A red circle 2 points to the 'Duties' tab. A red circle 3 points to the 'Create new' button. A red circle 4 points to the 'Name' field in the 'Enter new duty name' dialog, which contains 'New Custom Duty_001'. A red circle 5 points to the 'OK' button.

4. Create a new privilege by navigating to **Privileges > Create New**, enter the name "New Custom Privilege_001", then click **OK**.

The screenshot shows the 'Security configuration' window with the 'Privileges' tab selected. A red circle 1 points to the 'System administration > Security > Security configuration' breadcrumb. A red circle 2 points to the 'Privileges' tab. A red circle 3 points to the 'Create new' button. A red circle 4 points to the 'Name' field in the 'Enter new privilege name' dialog, which contains 'New Custom Privilege_001'. A red circle 5 points to the 'OK' button.

5. Go to **All Customers**, right-click on the **Name** column, select **Form Information**, and note the **Form Name: CustTable**

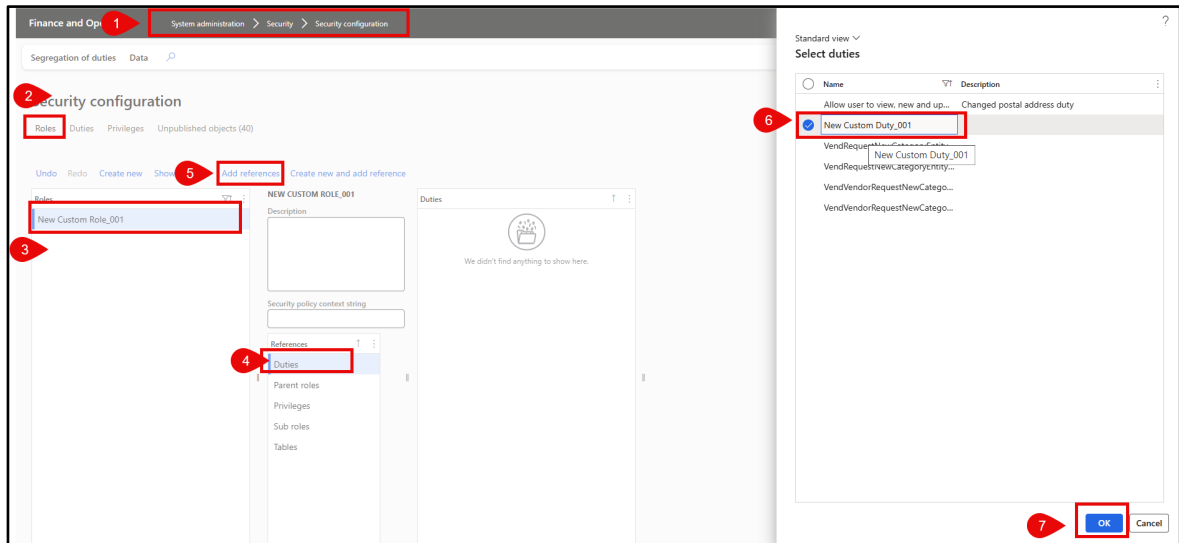
The screenshot shows the 'All customers' window. A red circle 1 points to the 'Accounts receivable > Customers > All customers' breadcrumb. A red circle 2 points to the 'Name' column in the customer list. A red circle 3 points to the 'Form information' option in the context menu. The 'Form Name: CustTable' is visible in the menu.

Account	Name	Invoice account	Customer group	Currency	Telephone
0000001	Frank the Cat	140	1	TRY	
0000001	One-time customer	80		USD	
0000002	demo 1			TZS	
0000003	One-time customer			USD	
0000004	One-time customer			HUF	
0000005	One-time customer			USD	
000012	HKI			USD	

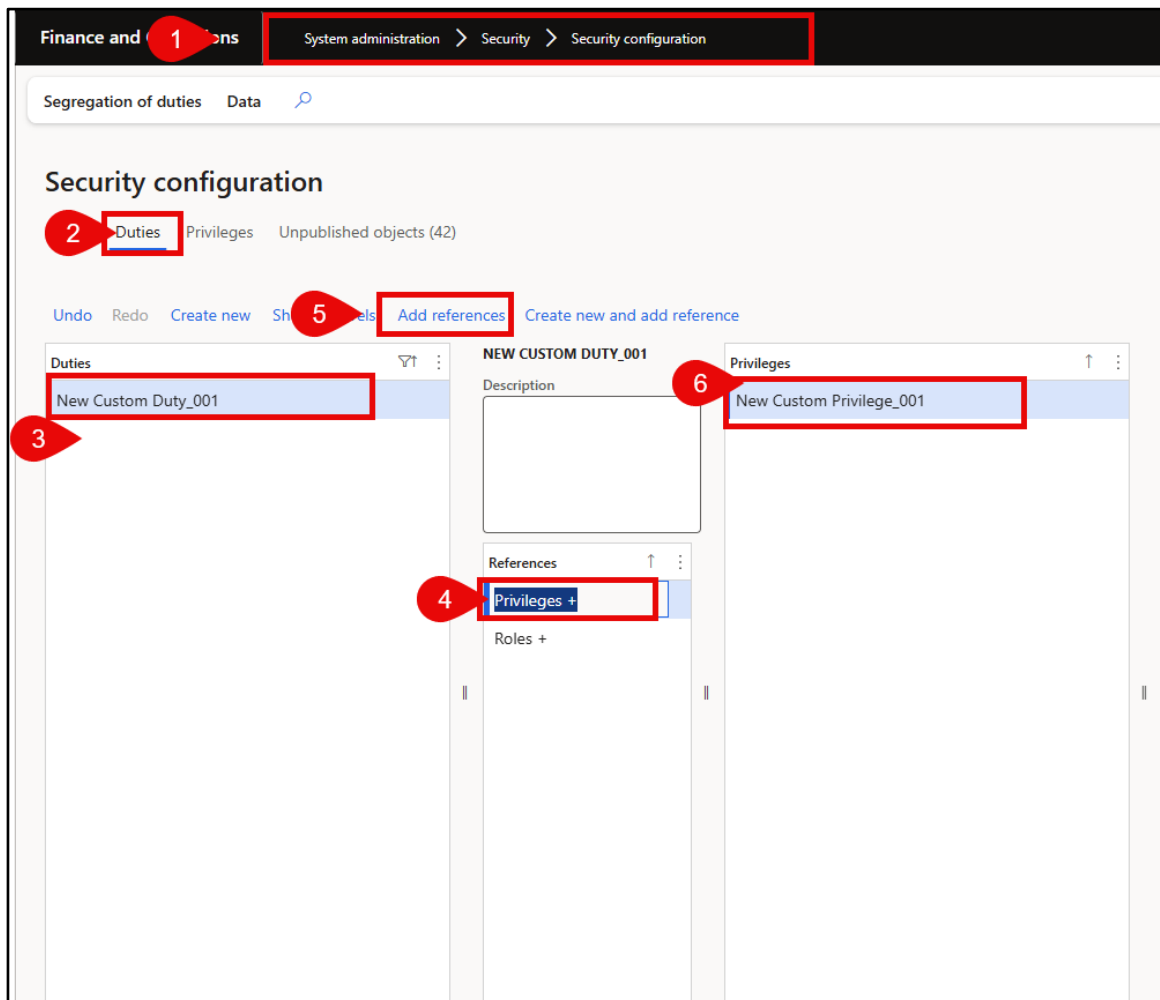
- Copy the **Menu Item Name** from the form information; it should be **CustTableListPage**.

- Go to **Privileges**, select the newly created privilege, then under **Display Menu Item**, click **Add Reference** and use the **Filter** to search for the menu item.

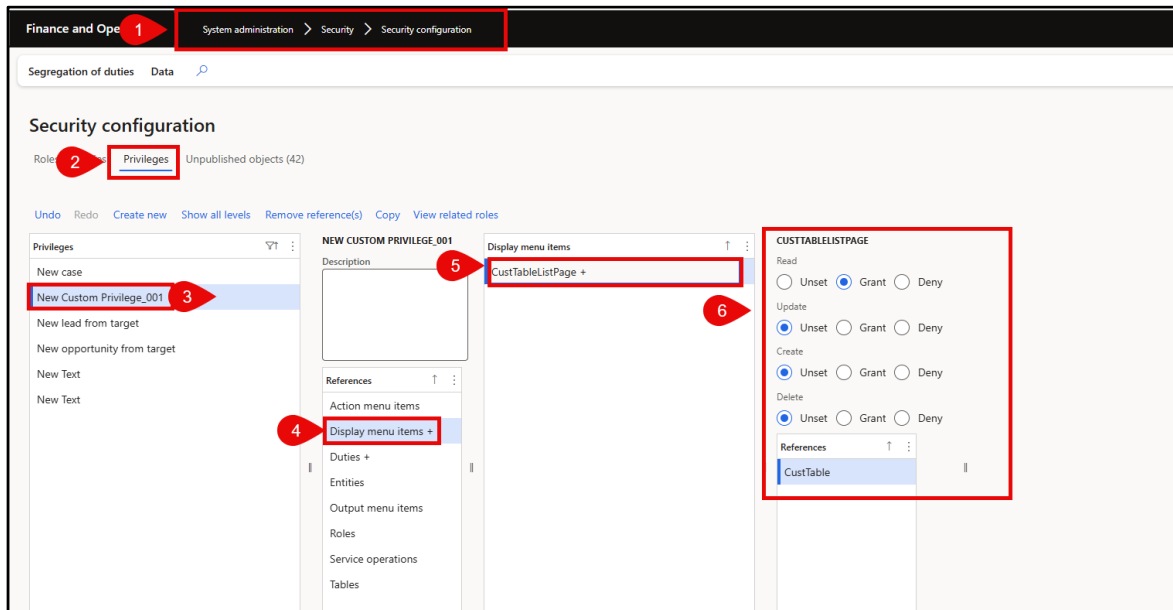
8. Now Go back to **Roles**, select the new role "**New Custom Role_001**", go to **Duties**, click **Add Reference**, filter and select "**New Custom Duty_001**", then click **OK**.



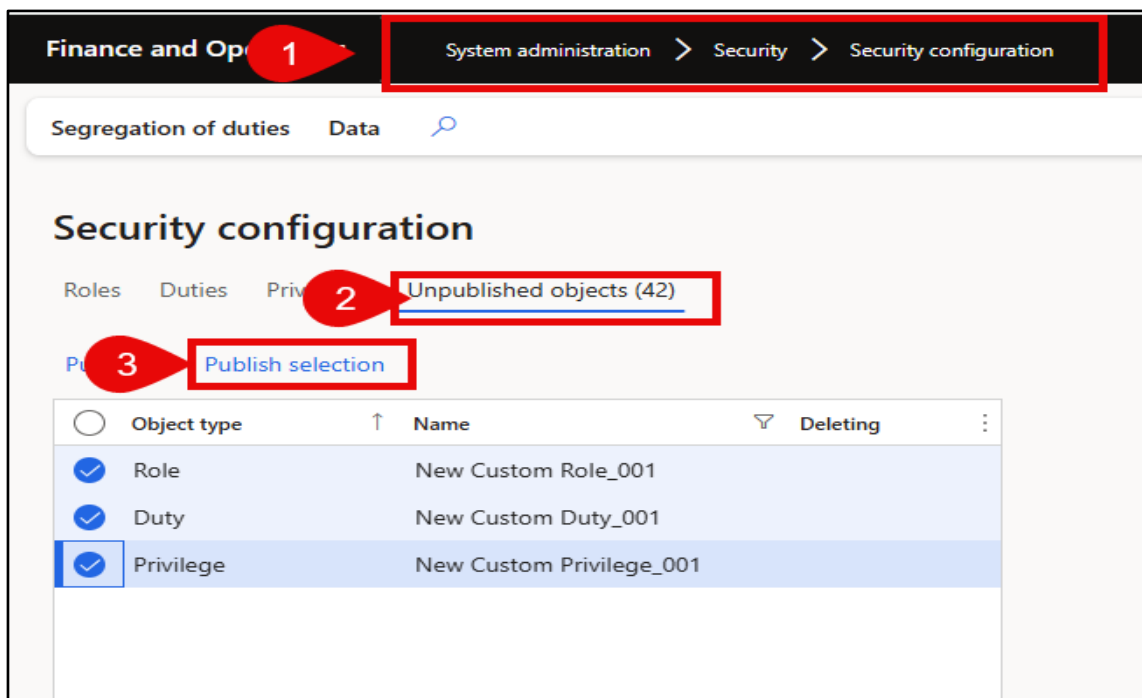
9. Go to **Duties**, select the created duty, navigate to **Privileges**, click **Add Reference**, find and select "**New Custom Privilege_001**", then click **OK**.



Now you can verify that **Privileges** are assigned to the **Duty**, and the **Duty** is assigned to the **Role**. You will also see that **CustTableListPage** has only **one permission granted** under the assigned privilege



10. After completing the above steps, go to 'Unpublished Objects', select all, and click **Publish**. Once published, the role is ready to be assigned by the system admin, and the assigned user will have **display-only** access to the **All Customers** view.



- After assigning the role, go to **Role > Assign Organization**, then choose whether to grant the permission to **all legal entities** or to a **specific one** by selecting the appropriate organization.

